

WEEK – 4 ASSIGNMENT

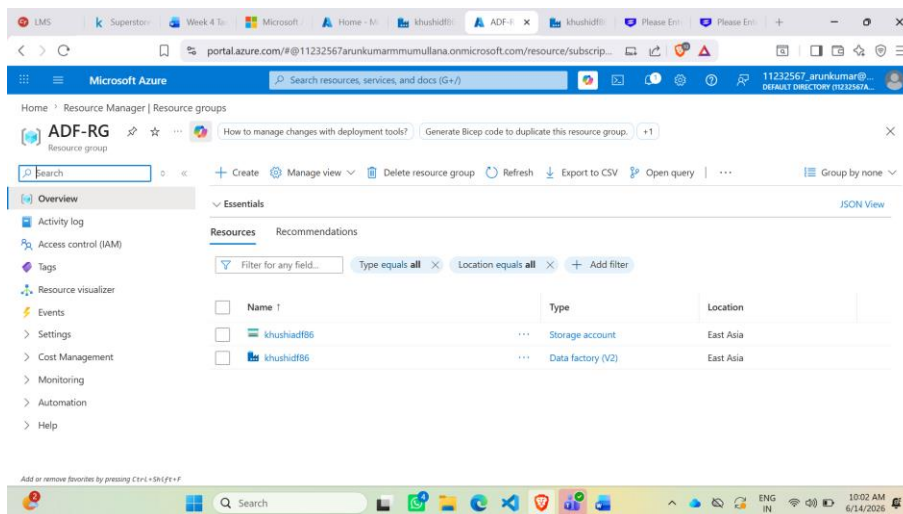
Title: Azure Cloud Fundamentals and Data Pipeline Implementation using ADF

Objective: To understand Azure cloud concepts and build a complete data pipeline using Storage Account and Azure Data Factory.

Assignment Tasks

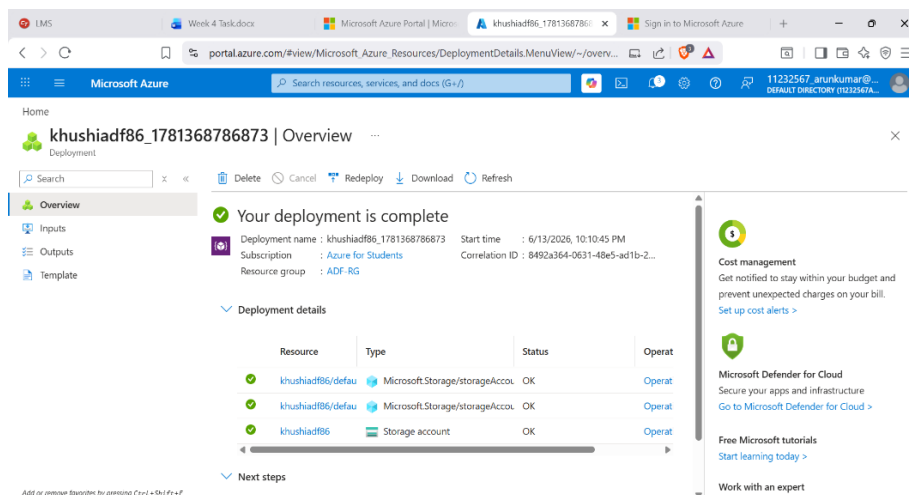
Task 1:

- Explore Azure Portal
- Create a Resource Group

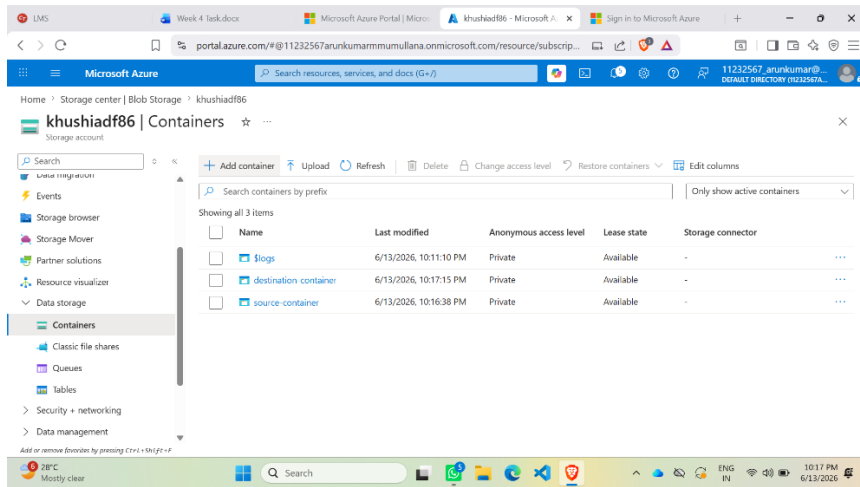


Task 2: Storage Setup

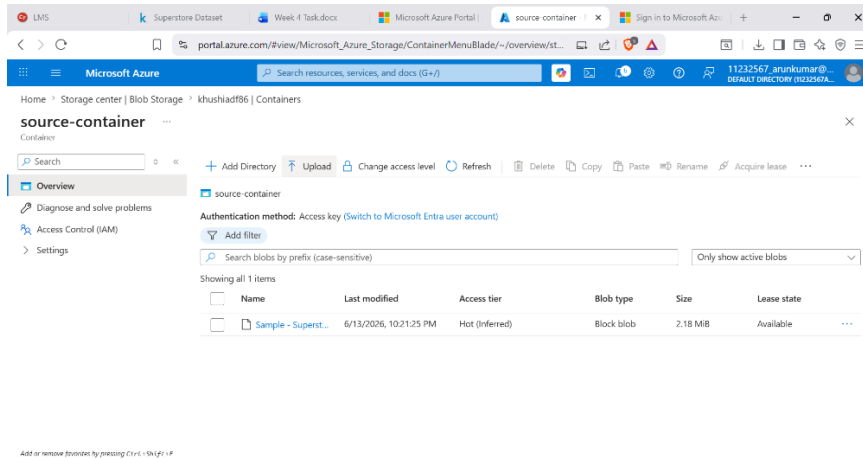
- Create a Storage Account



- Create a Blob Container

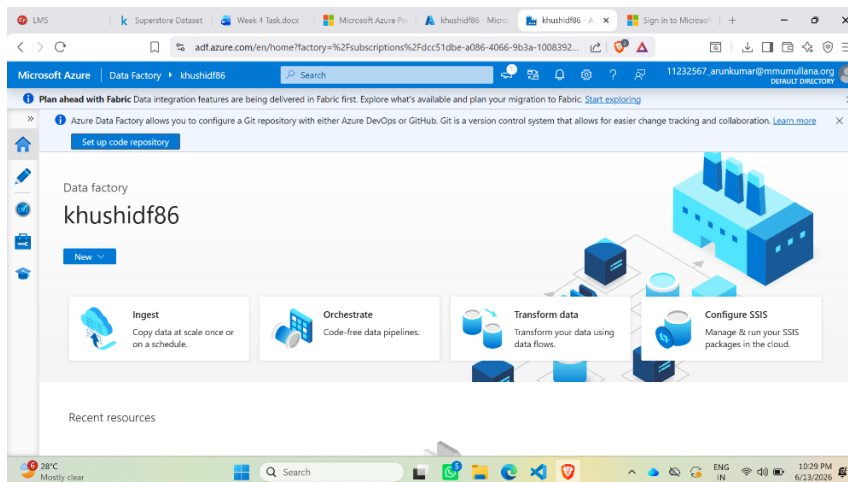


- Upload a CSV file

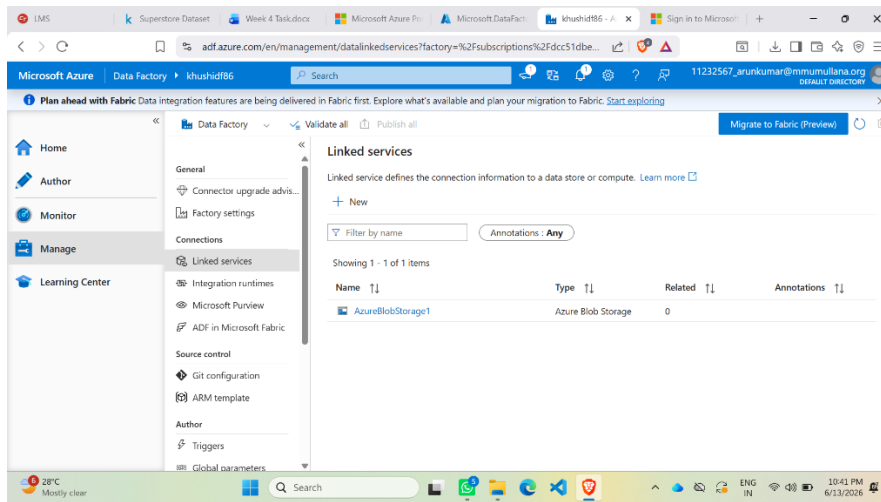


Task 3: ADF Basics

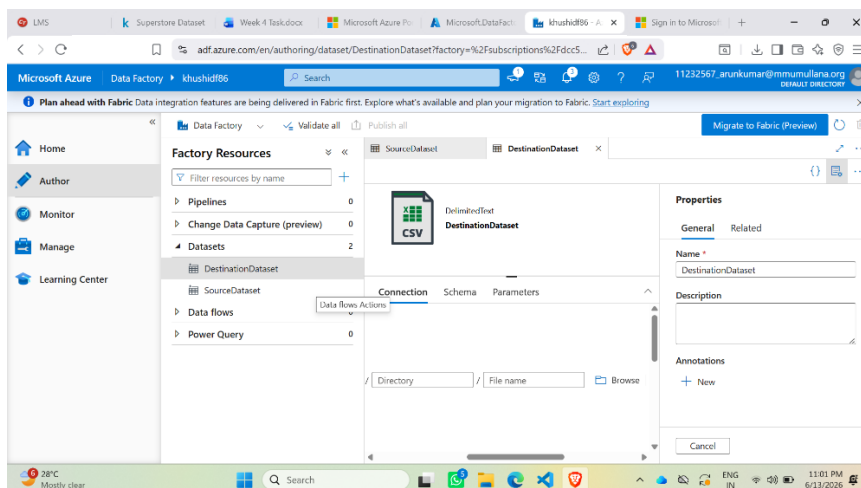
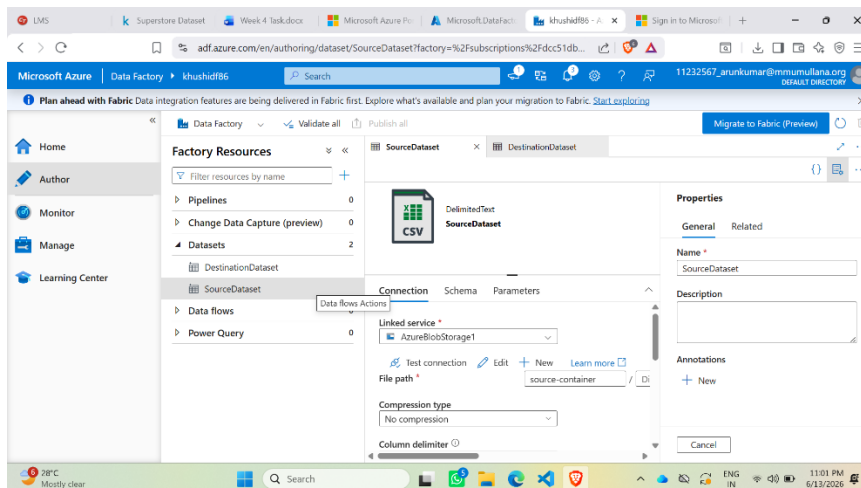
- Create Azure Data Factory



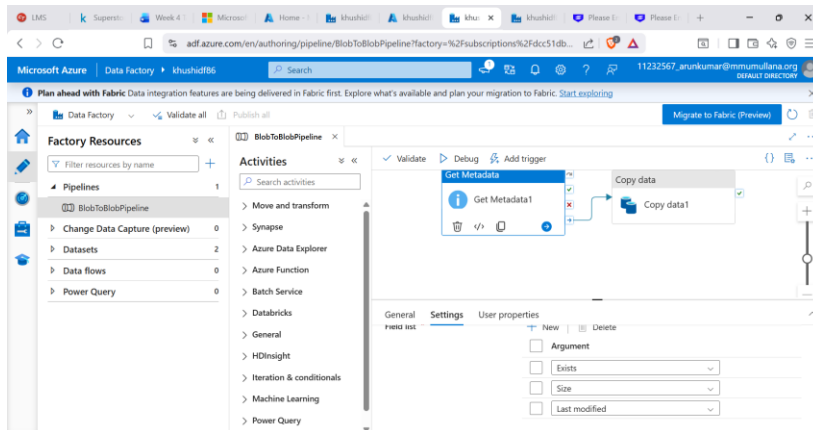
- Explore ADF UI (Author, Monitor, Manage)
- Create Linked Service (Blob Storage)



- Create datasets (source + destination)

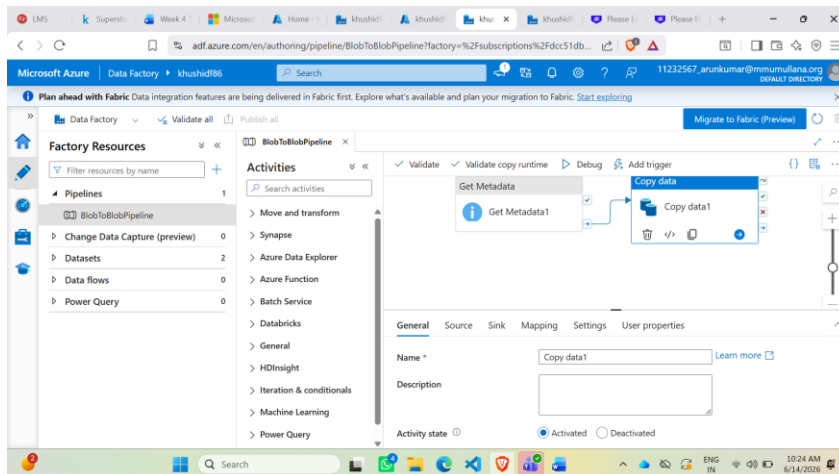


- Use Get Metadata activity

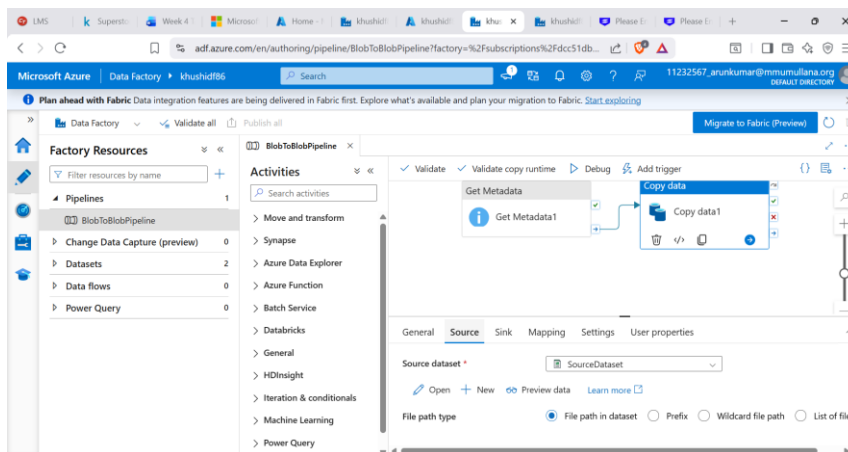


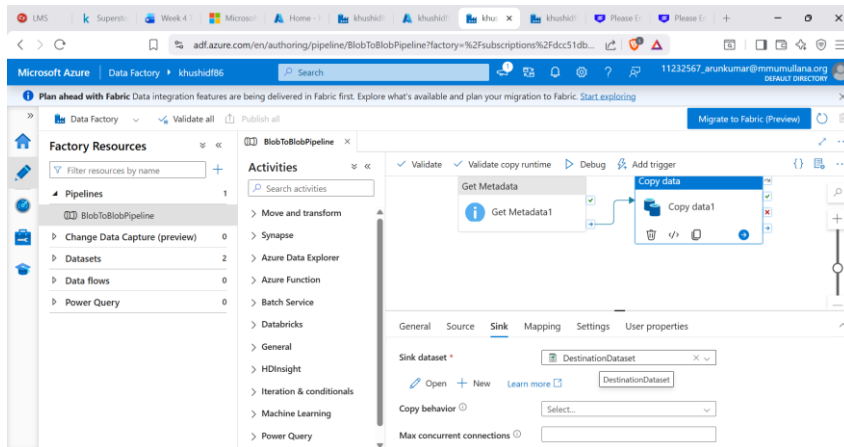
Task 4: Pipeline Development

- Create pipeline using Copy Data activity



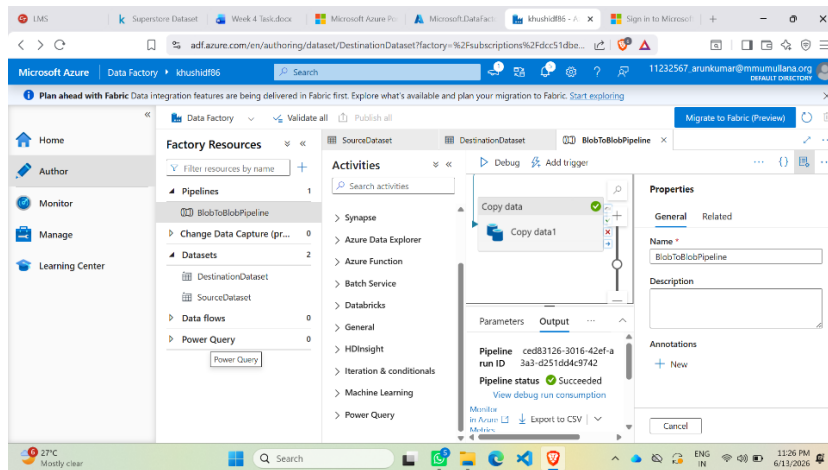
- Configure source and destination





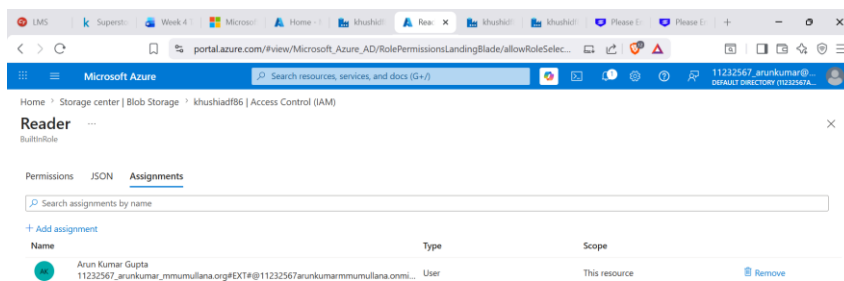
Task 5: Pipeline Execution

- Run pipeline using Debug/Trigger

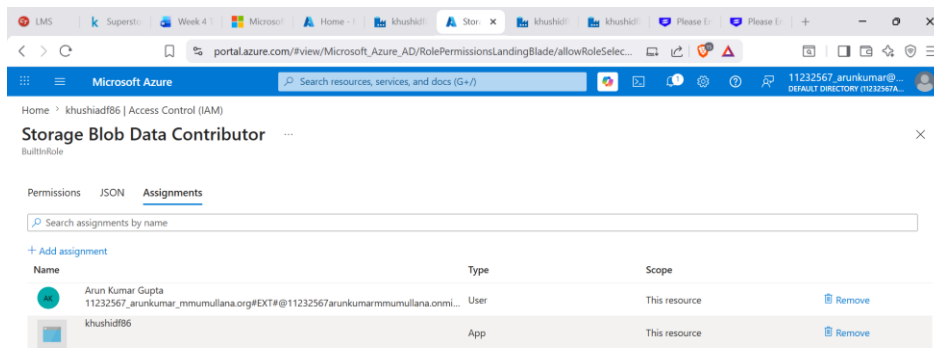


Task 6: IAM Roles

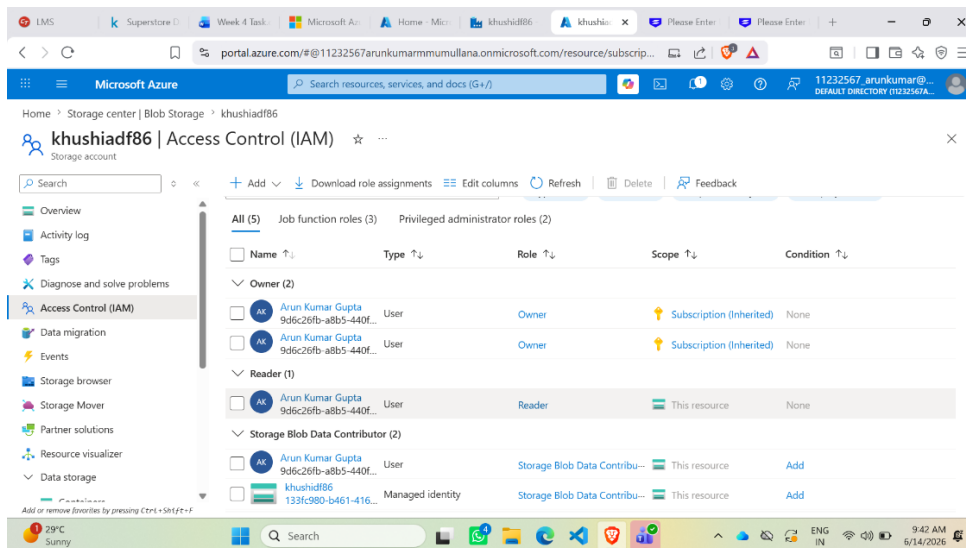
- Assign roles:
- Reader



- Contributor



- Provide access to ADF for Storage



Mini Project (Final Task)

Problem Statement: Build a complete pipeline that reads a CSV file from Blob Storage and processes it using Azure Data Factory.

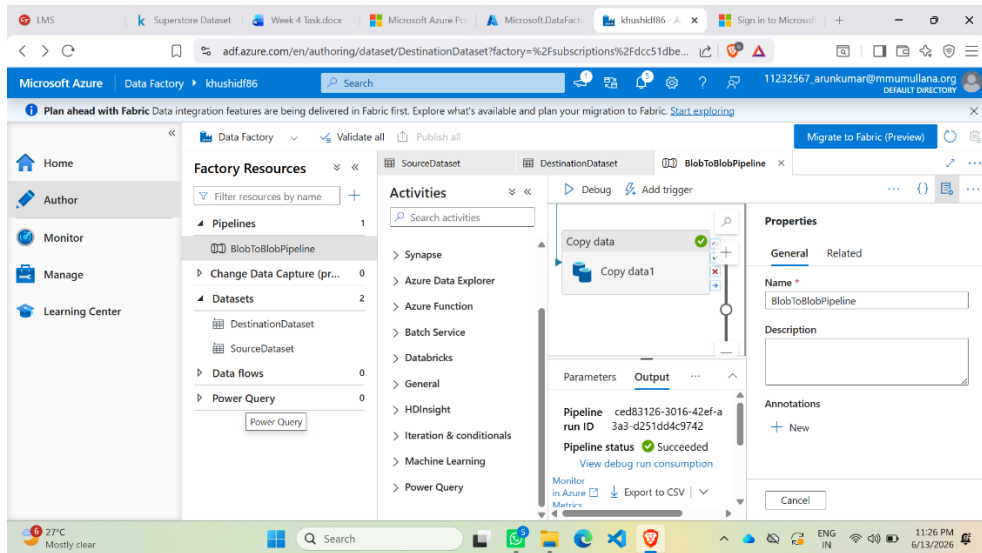
Requirements:

- Source: CSV file in Blob
- Use: Linked Service + Dataset + Pipeline
- Process: Copy Data + Metadata check

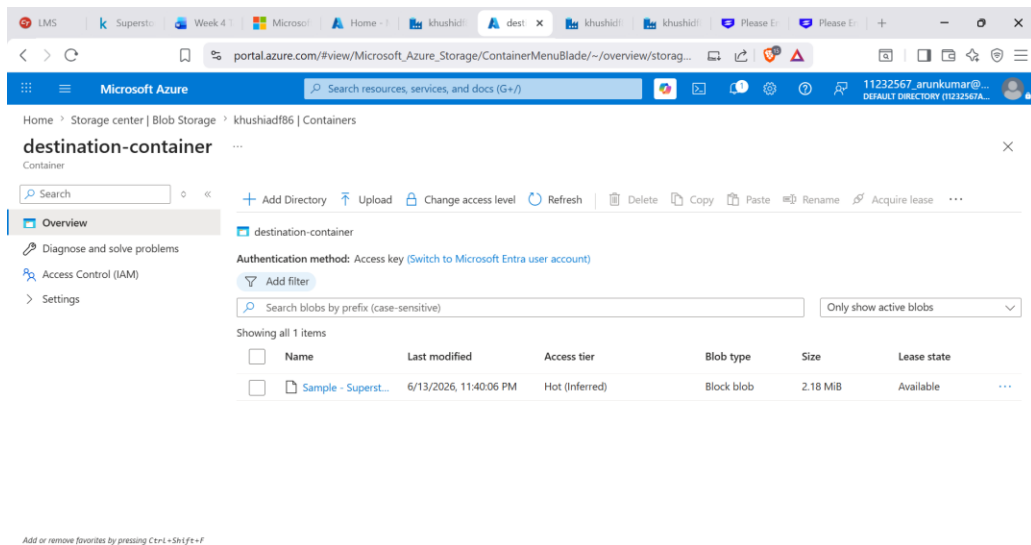
- Destination: New file/location

Expected Output:

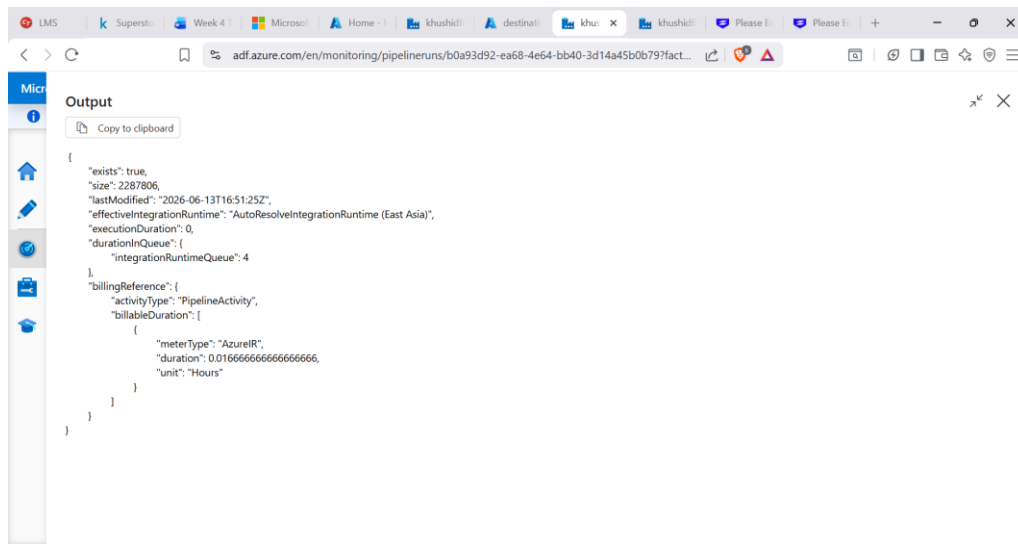
- Pipeline executed successfully



- Data copied to destination



- Metadata validated



BRIEF SUMMARY

This project involved building a simple data pipeline using Azure Data Factory and Azure Blob Storage. A Resource Group, Storage Account, and Blob containers were created, and a Superstore CSV file was uploaded to the source container. Azure Data Factory was configured with Linked Services and datasets, and a pipeline was developed using Get Metadata to validate the file and Copy Data to transfer it to the destination container. The pipeline was executed successfully and monitored through ADF. IAM roles were also assigned to enable secure access between Azure Data Factory and Azure Storage. This project provided practical experience with Azure cloud services, data movement, and pipeline monitoring.